

Advanced Torque Controller

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Lecture #6
Controller Design for Wind Turbines and Wind Farms

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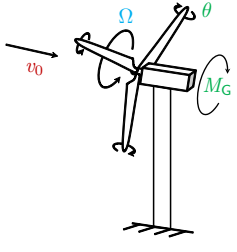
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Schedule

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Advanced generator torque controller



Motivation

- Generator torque is one of the two main control inputs.
- Baseline controller can be implemented in a simple form.
- Baseline controller performance and tuning options are limited.

$\rightarrow$ More advanced methods should get better results!

Main questions

- How can we design an advanced generator torque controller?
- How can we tune the advanced torque controller?

Contents

1. Recapitulation of the Torque Controller

2. Advanced Torque Controller Design

3. Conclusion and Learning Objectives

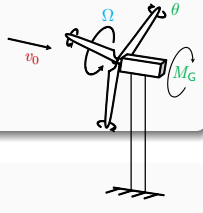


Controller design model

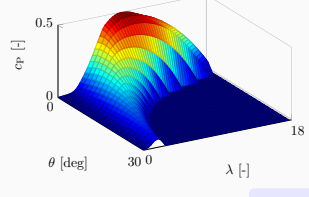
Structural dynamics

rotor motion: $J\dot{\Omega} = M_\mathrm{a} - M_\mathrm{G} r_\mathrm{GB}$

generator: $P_\mathrm{el} = M_\mathrm{G}\Omega r_\mathrm{GB}\eta_\mathrm{el}$



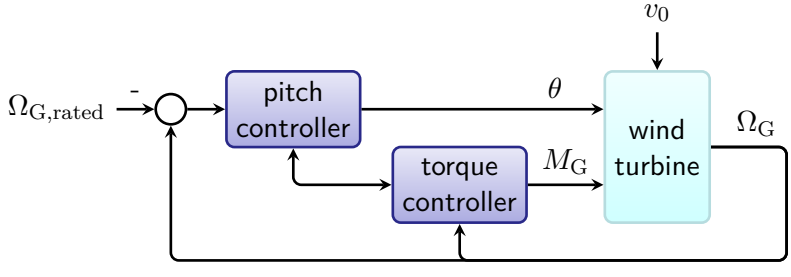
Aerodynamics



torque: $M_\mathrm{a} = \frac{1}{2}\rho\pi R^2 \frac{c_\mathrm{P}(\lambda,\theta)}{\Omega} v_0^3$

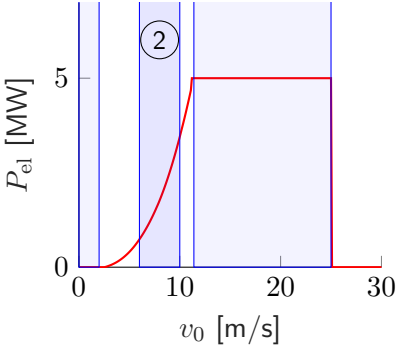
with tip speed ratio $\lambda = \frac{\Omega R}{v_0}$.

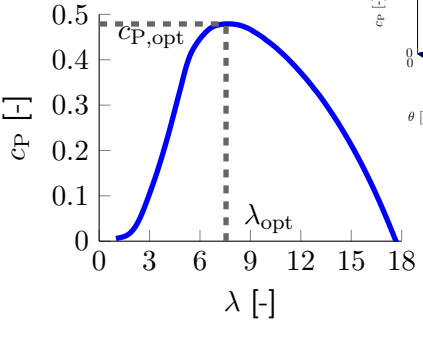
Baseline wind turbine controller

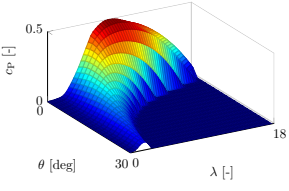


- Torque controller optimizes power below and maintains rated power above rated wind speed by adjusting generator torque M_{G} based on generator speed Ω_{G}.
- Pitch control maintains rated generator speed $\Omega_{\mathrm{G,rated}}$ by adjusting blade pitch angle θ.
- Interactions for the transition between above and below rated wind are very important!

Strategy baseline torque controller



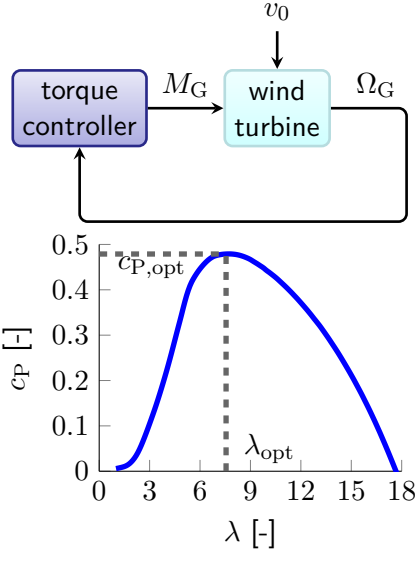




$\leftarrow \theta_{\mathrm{opt}} = 0\,\mathrm{deg}$

- Torque controller in Region 2 aims to maximize $P_{\mathrm{a}} = \frac{1}{2}\rho\pi R^2 c_{\mathrm{P}}(\lambda, \theta) v_0^3$.
- Optimum $c_{\mathrm{P,opt}}$ is reached for $\lambda = \lambda_{\mathrm{opt}}$ and $\theta_{\mathrm{opt}} = 0\,\mathrm{deg}$.
- Rotor-effective wind speed v_0 is not measurable.
- Indirect speed control: Feedback $M_{\mathrm{G}}(\Omega_{\mathrm{G}})$ can be designed such that $\lambda \to \lambda_{\mathrm{opt}}$.

Optimal torque control



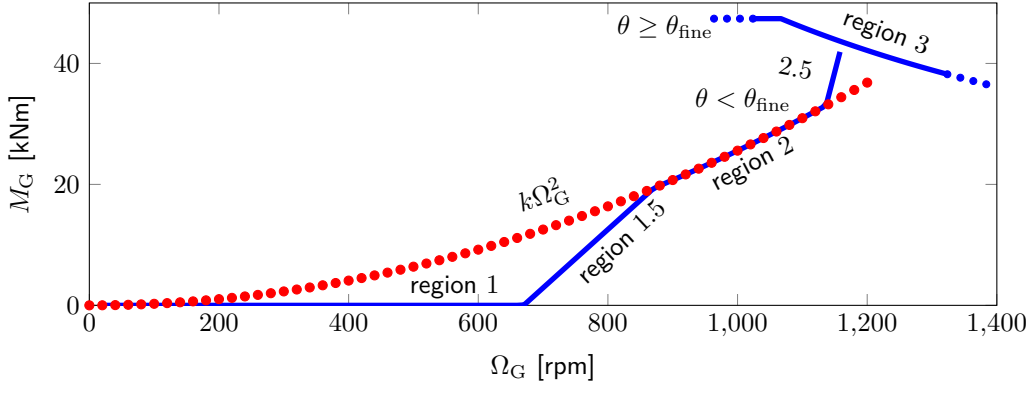
Optimal feedback

$$J\dot{\Omega} = M_\mathrm{a} - M_\mathrm{G} r_\mathrm{GB} \stackrel{!}{=} 0$$

$M_\mathrm{a} = \frac{1}{2}\rho\pi R^2 \frac{c_\mathrm{P,opt}}{\Omega} v_0^{\,3}$, $\Omega = \frac{\Omega R}{\lambda_\mathrm{opt}}$, $\Omega = \frac{\Omega_\mathrm{G}}{r_\mathrm{GB}}$

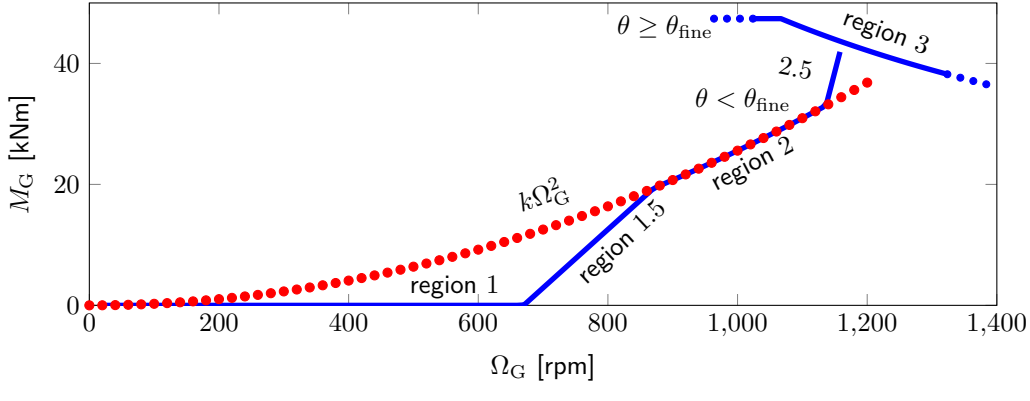
$$\rightarrow \quad M_\mathrm{G} = \underbrace{\frac{1}{2}\rho\pi R^5 \frac{c_\mathrm{P,opt}}{\lambda_\mathrm{opt}^3 r_\mathrm{GB}^3}}_{k} \Omega_\mathrm{G}^2$$

Baseline torque controller: nonlinear state feedback



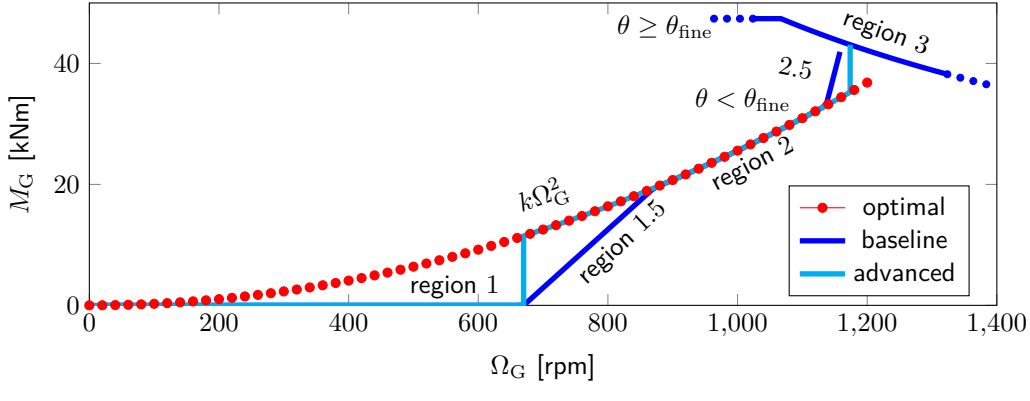
- Region 2: $k\ \Omega_{\text{G}}^2$ brings turbine to chosen power coefficient.
- Region 1.5 and 2.5: linear functions for transition, Region 3: onshore usually constant power
- Fine pitch value θ_{fine} to determine whether Region 3 or 2.5 should be applied

Limitations of baseline torque controller



- With ramps, optimal curve is reached later and left earlier compared to vertical lines.
- Dynamics on ramps can only be tuned indirectly by steepness.
- Switching due to fine pitch usually results in high loads.

Goals of advanced torque controller



- Optimal curve is reached earlier and left later compared to baseline torque controller.
- Dynamics on vertical lines (Regions 1.5 and 2.5) are tunable.
- Interaction with PI pitch controller for low loads and increased energy

Strategy advanced torque controller

- Torque PI controller for Region 1.5 and 2.5
- Torque PI controller parameters with closed-loop shaping similar to pitch PI controller
- Torque controller in Regions 2 and 3 by adjustment of limits
- Interaction with pitch controller in Region 2.5 necessary to avoid competing interactions
- Compromise between energy production, reduction of structural load, speed regulation, and pitch activity

Closed-loop shaping 1/5

Basic idea [1]

- Combining linearized 1 DOF model and PI controller results in a 2nd order linear model.
- Input is rotor effective wind v_0, output is generator speed Ω_G.
- Dynamics of the closed loop can be shaped by the parameters of the PI controller.
- Usually no gain scheduling is necessary.

Procedure

1. Linearize 1 DOF model at central operation point
2. Combine linearized 1 DOF model with PI controller to 2nd order linear model
3. Calculate parameters of PI controller at central operation point with desired dynamics

Closed-loop shaping 2/5

1 DOF model with $\theta = \text{const.}$

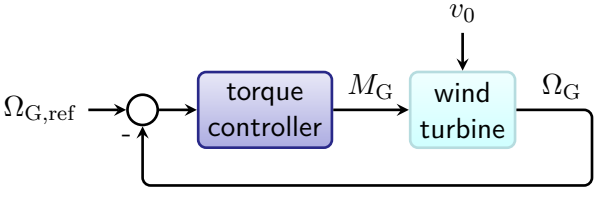
$\dot{\Omega} = \frac{M_\text{a}(\Omega, v_0)}{J} - \frac{M_\text{G} r_\text{GB}}{J} = f(\Omega, v_0, M_\text{G})$

1. Linearize 1 DOF model at central operation point

$$\Delta\dot{\Omega} = \underbrace{\frac{\partial f}{\partial \Omega}\Big|_\text{OP}}_{a} \Delta\Omega + \underbrace{\frac{\partial f}{\partial M_\text{G}}\Big|_\text{OP}}_{b_1} \Delta M_\text{G} + \underbrace{\frac{\partial f}{\partial v_0}\Big|_\text{OP}}_{b_2} \Delta v_0 \tag{1}$$

$$\Delta\Omega_\text{G} = \underbrace{r_\text{GB}}_{c} \Delta\Omega \tag{2}$$

Closed-loop shaping 3/5



PI controller

$$\Delta M_{\mathrm{G}} = K_{\mathrm{P}} \Delta\Omega_{\mathrm{G}} + K_{\mathrm{I}} \int_0^t \Delta\Omega_{\mathrm{G}} \mathrm{d}\tau \tag{3}$$

2. Combine linearized 1 DOF model with PI controller to 2nd order linear model

with (1)-(3) and $\dot{x} = \Delta\Omega$, $y = \Delta\Omega_{\mathrm{G}}$, and $u = \Delta v_0$:

$$\ddot{x} + (-a - b_1 c K_{\mathrm{P}})\dot{x} + (-b_1 c K_{\mathrm{I}})x = b_2 u \tag{4}$$

$$y = c\dot{x} \tag{5}$$

Closed-loop shaping 4/5

3. Calculate parameters of PI controller at central operation point

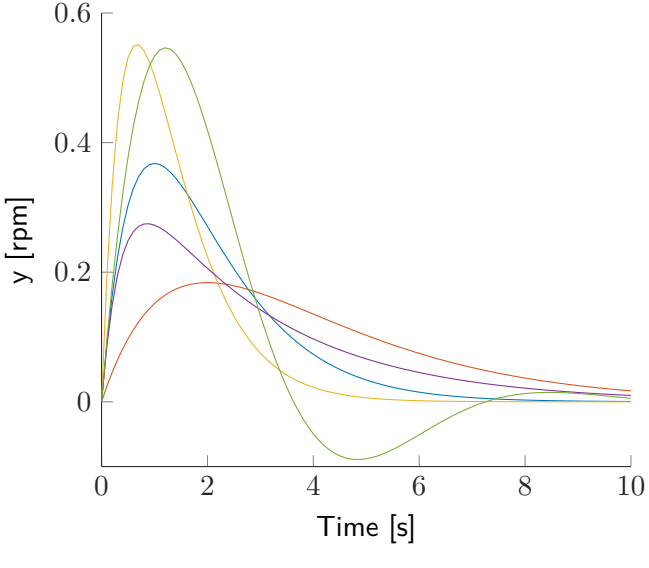
Parameter comparison: closed-loop transfer function vs. desired 2nd order linear model

$$\frac{cb_2\, s}{s^2 + (-a - b_1 c K_\mathrm{P})\, s + (-b_1 c K_\mathrm{I})} \stackrel{!}{=} \frac{G_0 \omega^2\, s}{s^2 + 2D\omega\, s + \omega^2}$$

Solution

Amplification factor:	G_0	$= \frac{cb_2}{\omega^2}$	with	Desired damping ratio:	D
Proportional gain:	K_P	$= -\frac{2D\omega + a}{b_1 c}$		Desired angular frequency:	ω
Integral gain:	K_I	$= -\frac{\omega^2}{b_1 c}$			

Closed-loop shaping 5/5



Response to $1\,\mathrm{m/s}$ wind steps

$G_0 = 1\,\mathrm{rpm/(m/s)}$

- $D = 1.0,\ \omega = 1.0\,\mathrm{rad/s}$
- $D = 1.0,\ \omega = 0.5\,\mathrm{rad/s}$
- $D = 1.0,\ \omega = 1.5\,\mathrm{rad/s}$
- $D = 1.5,\ \omega = 1.0\,\mathrm{rad/s}$
- $D = 0.5,\ \omega = 1.0\,\mathrm{rad/s}$

Reference values and limits

Reference values $\Omega_{\mathrm{G,ref}}$

- In Region 1.5: lowest generator speed $\Omega_{\mathrm{G},1.5}$, usually to avoid interaction of the 3P frequency with the tower eigenfrequency
- In Region 2.5: $\Omega_{\mathrm{G},2.5} = \Omega_{\mathrm{G,rated}}$
- Switch from $\Omega_{\mathrm{G},1.5}$ to $\Omega_{\mathrm{G},2.5}$ if the measured generator speed $\Omega_{\mathrm{G}} \geq \Omega_{\mathrm{G,R2switch}} = \frac{1}{2}(\Omega_{\mathrm{G},1.5} + \Omega_{\mathrm{G},2.5})$

Lower bound $M_{\mathrm{G,lb}}$ and upper bound $M_{\mathrm{G,ub}}$

Limits depend on measured generator speed Ω_{G} and are used together with anti-windup.

$$\Omega_{\mathrm{G}} < \Omega_{\mathrm{G,R2switch}}: \quad M_{\mathrm{G,lb}} = 0, \quad M_{\mathrm{G,ub}} = k\Omega_{\mathrm{G}}^2$$

$$\Omega_{\mathrm{G}} \geq \Omega_{\mathrm{G,R2switch}}: \quad M_{\mathrm{G,lb}} = k\Omega_{\mathrm{G}}^2, \quad M_{\mathrm{G,ub}} = \min(M_{\mathrm{G,rated}}\Omega_{\mathrm{G,rated}}/\Omega_{\mathrm{G}}, M_{\mathrm{G,max}})$$

Methods for pitch and torque PI interactions [2]

Double PI

- The difference to the rated power is multiplied with a proportional and integral gain and added to the proportional and integral gain for the speed error (only one integrator).
- Problem: not clear how to tune

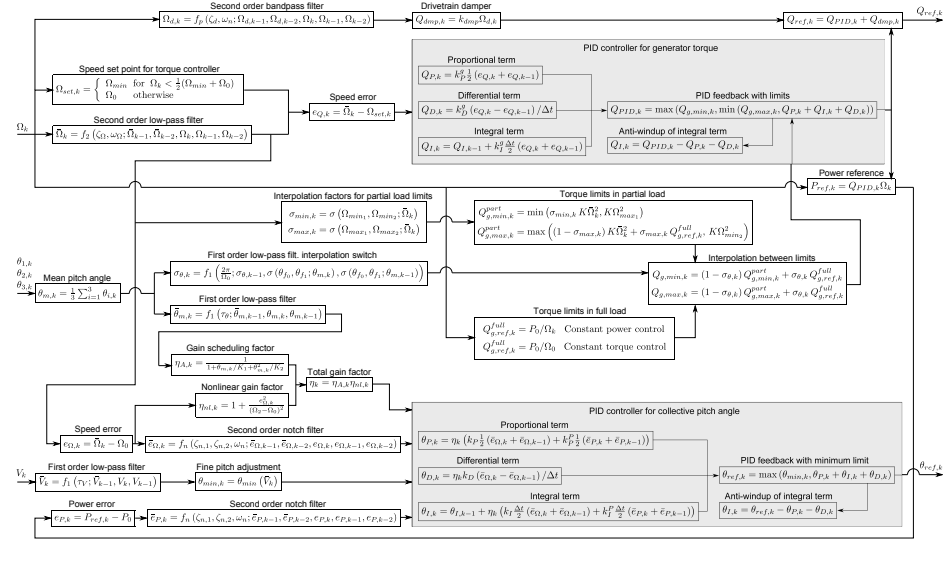
State machine

- Based on conditions for the pitch angle etc., a state machine is defined including hysteresis to avoid both controllers reacting at the same time.
- Problem: Very nonlinear, controller can be caught in the "wrong" mode.

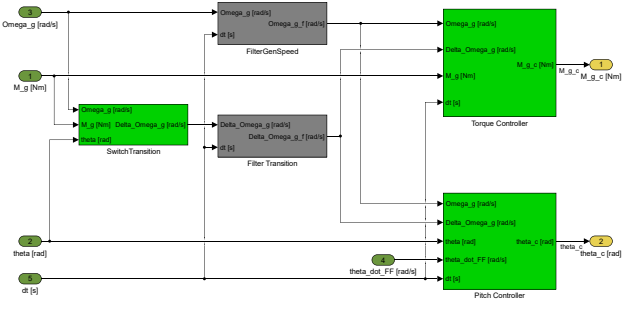
Set-point-fading

- An additional term (bias) $\Delta\Omega_{\mathrm{G}}$ is added/subtracted from the reference speed, such that only one of the two PI controllers has the rated generator speed as reference value.
- Relatively easy and intuitive to tune

Double PI [3]

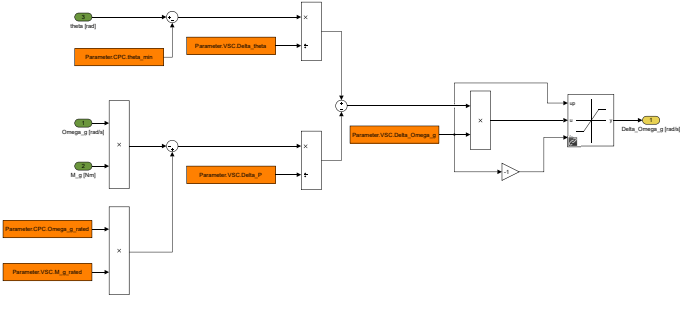

Second order bandpass filter
$\Omega_{d,k} = f_p(\zeta_d, \omega_n; \Omega_{d,k-1}, \Omega_{d,k-2}, \Omega_k, \Omega_{k-1}, \Omega_{k-2})$
Drivetrain damper
$Q_{dmp,k} = k_{dmp}\Omega_{d,k}$
$Q_{ref,k}$
$Q_{ref,k} = Q_{PID,k} + Q_{dmp,k}$
PID controller for generator torque
Proportional term
$Q_{P,k} = k_P^g \frac{1}{2}(e_{Q,k} + e_{Q,k-1})$
Differential term
$Q_{D,k} = k_D^g (e_{Q,k} - e_{Q,k-1}) / \Delta t$
Integral term
$Q_{I,k} = Q_{I,k-1} + k_I^g \frac{\Delta t}{2}(e_{Q,k} + e_{Q,k-1})$
PID feedback with limits
$Q_{PID,k} = \max(Q_{g,min,k}, \min(Q_{g,max,k}, Q_{P,k} + Q_{I,k} + Q_{D,k}))$
Anti-windup of integral term
$Q_{I,k} = Q_{PID,k} - Q_{P,k} - Q_{D,k}$
Speed set point for torque controller
$\Omega_{set,k} = \begin{cases} \Omega_{min} & \text{for } \Omega_k < \frac{1}{2}(\Omega_{min} + \Omega_0) \\ \Omega_0 & \text{otherwise} \end{cases}$
Speed error
$e_{Q,k} = \bar{\Omega}_k - \Omega_{set,k}$
Ω_k
Second order low-pass filter
$\bar{\Omega}_k = f_2(\zeta_\Omega, \omega_n; \bar{\Omega}_{k-1}, \bar{\Omega}_{k-2}, \Omega_k, \Omega_{k-1}, \Omega_{k-2})$
Power reference
$P_{ref,k} = Q_{PID,k}\Omega_k$
Interpolation factors for partial load limits
$\sigma_{min,k} = \sigma(\Omega_{min_1}, \Omega_{min_2}; \bar{\Omega}_k)$
$\sigma_{max,k} = \sigma(\Omega_{max_1}, \Omega_{max_2}; \bar{\Omega}_k)$
Torque limits in partial load
$Q_{g,min,k}^{part} = \min(\sigma_{min,k} K\bar{\Omega}_k^2, K\Omega_{min_2}^2)$
$Q_{g,max,k}^{part} = \max\left((1-\sigma_{max,k}) K\bar{\Omega}_k^2 + \sigma_{max,k} Q_{g,ref,k}^{full}, K\Omega_{min_2}^2\right)$
$\theta_{1,k}$
$\theta_{2,k}$
$\theta_{3,k}$
Mean pitch angle
$\bar{\theta}_{m,k} = \frac{1}{3}\sum_{i=1}^{3} \theta_{i,k}$
First order low-pass filt. interpolation switch
$\sigma_{\theta,k} = f_1\left(\frac{2\pi}{\Omega_0}; \sigma_{\theta,k-1}, \sigma(\theta_{f_0}, \theta_{f_1}; \bar{\theta}_{m,k}), \sigma(\theta_{f_0}, \theta_{f_1}; \bar{\theta}_{m,k-1})\right)$
Interpolation between limits
$Q_{g,min,k} = (1-\sigma_{\theta,k}) Q_{g,min,k}^{part} + \sigma_{\theta,k} Q_{g,ref,k}^{full}$
$Q_{g,max,k} = (1-\sigma_{\theta,k}) Q_{g,max,k}^{part} + \sigma_{\theta,k} Q_{g,ref,k}^{full}$
First order low-pass filter
$\bar{\theta}_{m,k} = f_1(\tau_\theta; \bar{\theta}_{m,k-1}, \theta_{m,k}, \theta_{m,k-1})$
Torque limits in full load
$Q_{g,ref,k}^{full} = P_0/\Omega_k$ Constant power control
$Q_{g,ref,k}^{full} = P_0/\Omega_0$ Constant torque control
Gain scheduling factor
$\eta_{A,k} = \frac{1}{1+\theta_{m,k}/K_1+\theta_{m,k}^2/K_2}$
Total gain factor
$\eta_k = \eta_{A,k}\eta_{nl,k}$
Nonlinear gain factor
$\eta_{nl,k} = 1 + \frac{c_0^\theta}{(\Omega_k - \Omega_0)^2}$
PID controller for collective pitch angle
Proportional term
$\theta_{P,k} = \eta_k \left(k_P \frac{1}{2}(\bar{e}_{\Omega,k} + \bar{e}_{\Omega,k-1}) + k_P^{\rho} \frac{1}{2}(\bar{e}_{P,k} + \bar{e}_{P,k-1})\right)$
Differential term
$\theta_{D,k} = \eta_k k_D (\bar{e}_{\Omega,k} - \bar{e}_{\Omega,k-1}) / \Delta t$
Integral term
$\theta_{I,k} = \theta_{I,k-1} + \eta_k \left(k_I \frac{\Delta t}{2}(\bar{e}_{\Omega,k} + \bar{e}_{\Omega,k-1}) + k_I^{\rho} \frac{\Delta t}{2}(\bar{e}_{P,k} + \bar{e}_{P,k-1})\right)$
PID feedback with minimum limit
$\theta_{ref,k} = \max(\theta_{min,k}, \theta_{P,k} + \theta_{I,k} + \theta_{D,k})$
Anti-windup of integral term
$\theta_{I,k} = \theta_{ref,k} - \theta_{P,k} - \theta_{D,k}$
$\theta_{ref,k}$
Speed error
$\bar{e}_{\Omega,k} = \bar{\Omega}_k - \Omega_0$
Second order notch filter
$\bar{e}_{\Omega,k} = f_n(\zeta_{n,1}, \zeta_{n,2}, \omega_n; \bar{e}_{\Omega,k-1}, \bar{e}_{\Omega,k-2}, e_{\Omega,k}, e_{\Omega,k-1}, e_{\Omega,k-2})$
V_k
First order low-pass filter
$\bar{V}_k = f_1(\tau_V; \bar{V}_{k-1}, V_k, V_{k-1})$
Fine pitch adjustment
$\theta_{min,k} = \theta_{min}(\bar{V}_k)$
Power error
$e_{P,k} = P_{ref,k} - \bar{P}_0$
Second order notch filter
$\bar{e}_{P,k} = f_n(\zeta_{n,1}, \zeta_{n,2}, \omega_n; \bar{e}_{P,k-1}, \bar{e}_{P,k-2}, e_{P,k}, e_{P,k-1}, e_{P,k-2})$

Set-point-fading overview



- Close to industrial practice
- Modified version of [2]
- $\Delta\Omega_{\mathrm{G}}$ is low pass filtered
- Input to torque PI controller and pitch PI controller
- Developed at sowento, used at SWE, University of Stuttgart in several projects
- Can be combined with multi-variable lidar-based feedforward control [4]
- NREL is now using the sowento approach [5].

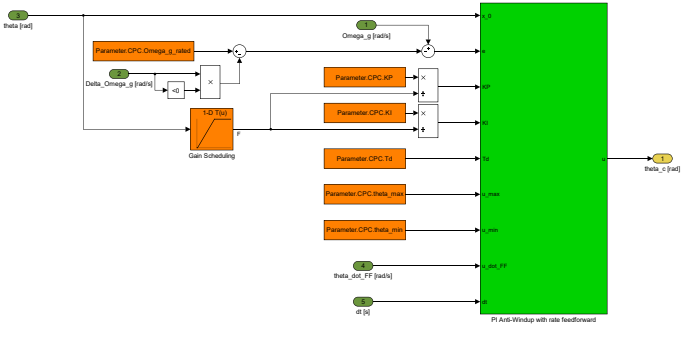
Switch transition



Transition

- Deviation from minimum pitch and deviation from rated power as proxy for deviation from rated wind
- Normalized by $\Delta\theta$ and ΔP
- Multiplied and saturated with maximum tolerated over-/under-speed $\Delta\Omega_{\mathrm{G,max}}$

Integration into pitch/torque controller



Pitch controller

If $\Delta\Omega_\mathrm{G} < 0$ then

$\Omega_\mathrm{G,ref} = \Omega_\mathrm{G,rated} - \Delta\Omega_\mathrm{G}$,

else

$\Omega_\mathrm{G,ref} = \Omega_\mathrm{G,rated}$

Torque controller

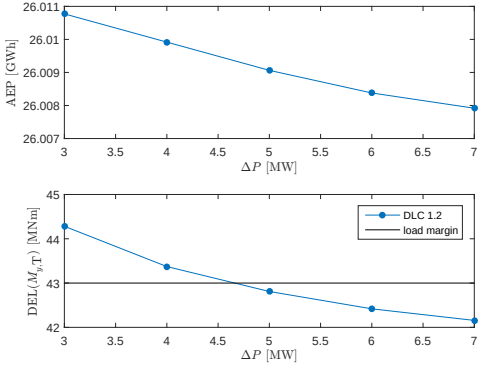
If $\Delta\Omega_\mathrm{G} > 0$ then

$\Omega_\mathrm{G,ref} = \Omega_\mathrm{G,rated} - \Delta\Omega_\mathrm{G}$,

else

$\Omega_\mathrm{G,ref} = \Omega_\mathrm{G,rated}$

Rule-of-thumb and fine tuning



Rule-of-thumb

- $\Delta\Omega_{\mathrm{G,max}}$ maximum allowed over-speed, e.g. 10 % of $\Omega_{\mathrm{G,rated}}$
- ΔP close to rated power
- $\Delta\theta$ close to the pitch angle at cut-out wind speed (e.g. 25 deg)

Fine tuning

- ΔP and $\Delta\theta$ can be optimized by brute-force optimization.
- Run a full DLC 1.2 [6] for fatigue load reduction
- Useful objective: maximize AEP without exceeding the load margins

Conclusion

Main questions

- How can we design an advanced generator torque controller?
- How can we tune the advanced torque controller?

PI controller with anti-windup

- Closed-loop shaping similar to pitch controller, usually no gain scheduling
- Adjustment of reference values and bounds
- Several methods for coupling with pitch PI controller available: set-point fading is easy to tune

Linear and nonlinear tuning

- Closed-loop shaping with similar values compared to pitch PI controller
- Coupling with pitch PI controller is very nonlinear $\rightarrow$ brute-force optimization.

Quick check on learning objectives

After this lecture you should be able to...

- ...linearize a reduced turbine model in Region 2.
- ...design a PI torque controller.
- ...describe how the limits and reference values for the torque PI are obtained.
- ...describe methods for coupling the two PI controllers.

References

[1] M. H. Hansen, A. Hansen, T. J. Larsen, S. Øye, P. Sørensen, and P. Fuglsang. *Control design for a pitch-regulated, variable speed wind turbine*. Tech. rep. Risoe-R No. 1500(EN). Forskningscenter Risoe, 2005. URL: https://orbit.dtu.dk/en/publications/control-design-for-a-pitch-regulated-variable-speed-wind-turbine.

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[3] M. Hansen and L. C. Henriksen. *Basic DTU Wind Energy controller*. E-Report 0028(E). DTU Wind Energy, 2013. URL: https://orbit.dtu.dk/en/publications/basic-dtu-wind-energy-controller.

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Please let me know if you have further questions!

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